

NEURAL SYSTEMS

DEVELOPMENTAL SYSTEMS

Phenotype

under selection

Network

circuit or regulatory

Cells

identity

Molecules

channels, genes,
enhancers

Stomatogastric

ganglion · crabs

Swim circuit

two nudibranchs

Cortex

human vs mouse

eve stripe 2

fruit fly species

Gap genes

two fly species

Vulva

two nematodes

pyloric rhythm

swimming

synaptic
strengths
(models)

role of Si1 in
the circuit

identified
neurons

homologous
neurons

matched cell
types

K⁺
conductances,
2–4×

gene
expression

stripe 2
expression

gap expression
pattern

vulval cell
fates

interaction
strengths

EGF vs Wnt
induction

precursor
cells

enhancer
binding sites

maternal
inputs

Ras activity
(cryptic)

 conserved between individuals or species  varies or turns over  not examined